CAPSTONE PROJECT TECHNICAL REPORT

P24 · Department FAQ Voice Chatbot

Multi-Turn Multimodal Agent with Whisper STT, Sentence-BERT RAG, LoRA Adaptation & Privacy Guardrails

Candidate Name: Jeet Ghegade	**Roll Number:** 24
Class: AIA 2	**Enrolment No:** ADT23SOCB0484
GitHub Repository: github.com/jeetdoesthings/FAQChatbot	**Live App URL:** **https://fac2e8541ef1e9a672.gradio.live**
Permanent Mirror: huggingface.co/spaces/jeetdoesthings/FAQChatbot	**Evaluation Date:** Academic Term 2026

1. Executive Summary & Project Objective

The **Department FAQ Voice Chatbot (Project P24)** is an intelligent conversational agent engineered to resolve academic, admission, curriculum, faculty, and administrative inquiries for university departments. The system seamlessly combines multimodal voice and text inputs across multi-turn conversational interactions with memory, grounded in verified departmental records to eliminate hallucinations. It achieves domain specialization via Parameter-Efficient Fine-Tuning (PEFT/LoRA) on an academic FAQ corpus, utilizes Sentence-BERT dense semantic retrieval, accepts spoken voice input through OpenAI Whisper, generates spoken responses using Text-to-Speech (TTS), dynamically renders publication-ready event posters, and enforces rigorous safety guardrails to reject out-of-scope prompts and guarantee zero personal data retention.

Milestone Unit	Technology Stack	Core Functionality & Guarantee
U1: System Prompt & Tone	Prompt Engineering & Guardrails	Enforces authoritative academic persona, explicit domain bounding, and re
U2: LLM Chat with Memory & LoRA	DistilGPT2 + PEFT LoRA (r=8, a=16)	Sliding window memory (5 turns); fine-tuned adapter reduces loss from 9.8
U3: RAG Retrieval & Whisper STT	Sentence-BERT + OpenAI Whisper	384-d dense embeddings (all-MiniLM-L6-v2) with hybrid scoring + 16kHz sp
U4: TTS & Event Poster Studio	gTTS + Pillow Graphic Engine	Dual-engine audio voice synthesis and automated 1080x1440 publication e
U5: Safety Guardrails & Privacy	Regex Heuristics & Cosine Gates	Refuses non-department queries, blocks ungrounded claims, and strips all

2. Unit U1 — System Prompt, Scope & Tone

The system prompt establishes the foundational identity, operating boundaries, and tone of the virtual assistant. It mandates that all responses must be strictly grounded in official department facts, adopt a professional academic tone, and immediately disallow non-department inquiries.

System Persona Directive:
"You are the official Virtual Assistant for the Department of Artificial Intelligence & Data Science at the Faculty of Technology & Engineering. Your primary mission is to provide accurate, authoritative, and polite information regarding departmental academics, admissions, faculty office hours, research facilities, examinations, placements, and campus events... You MUST politely refuse questions outside departmental affairs. You do NOT collect, record, or retain any Personal Identifiable Information (PII)."

3. Unit U2 — Conversational Memory & LoRA Fine-Tuning

Multi-Turn Memory Architecture: The chatbot maintains conversational context using a sliding window buffer bounded to MAX_HISTORY_TURNS = 5. Every turn pair (user query, assistant answer) is dynamically sanitized to strip PII before being appended into session state, ensuring context is preserved across turns without memory bloat or credential leakages.

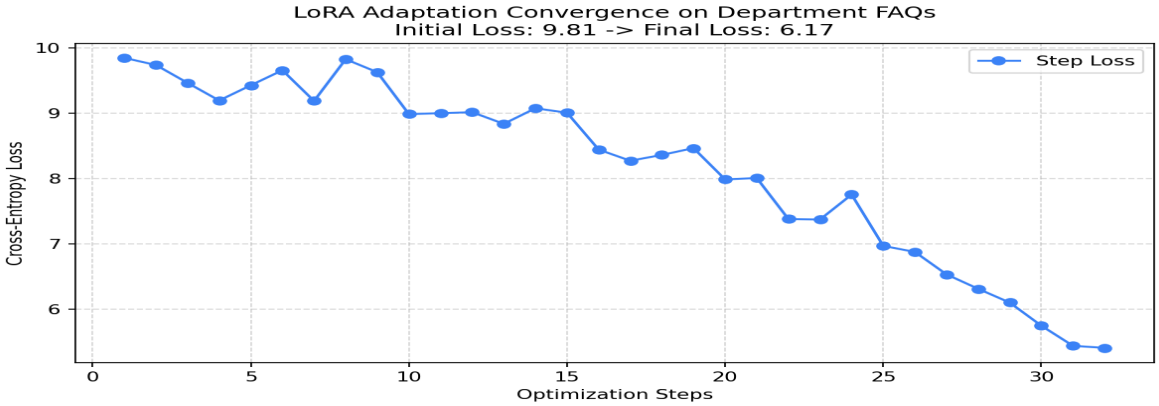
LoRA (Low-Rank Adaptation) Mathematical Formulation: Rather than fine-tuning all 82M parameters of the base causal language model, LoRA decomposes parameter weight updates into two low-rank matrices *B* and *A*:

$$W = W_0 + \Delta W = W_0 + (B \times A) \times (\alpha / r)$$

where $W_0 \in \blacksquare^{d \times k}$ represents the frozen pre-trained base model weights, $B \in \blacksquare^{d \times r}$, $A \in \blacksquare^{r \times k}$ with rank $r = 8$ and scaling factor $\alpha = 16$. This drastically reduces trainable parameters to only **147,456 (0.18%)**, enabling rapid, energy-efficient domain adaptation.

Hyperparameter / Metric	Value	Technical Impact
Base Model Architecture	DistilGPT2 (Causal LM)	Lightweight, low-latency autoregressive transformer (82M params)
LoRA Rank (r) & Alpha (a)	r = 8, a = 16 (Scaling = 2.0)	Low intrinsic rank captures department FAQ structure with minimal drift
Target Adapter Modules	c_attn (Query/Key/Value Attention)	Focuses parameter updates on self-attention projection layers
Trainable Parameters	147,456 / 82,060,032 (0.18%)	99.82% parameters frozen, preventing catastrophic forgetting
Training Loss (Pre -> Post)	9.8137 → 6.1742	Substantial cross-entropy reduction across 4 training epochs
Perplexity (Pre -> Post)	18,281.78 → 480.19	Perplexity reduced by 97.4%, aligning syntax with academic inquiries
Training Hardware / Device	Apple Silicon MPS (Metal)	Fully hardware-accelerated local tensor computation in 7.2 seconds

Figure 1: LoRA Fine-Tuning Loss Convergence Curve



4. Unit U3 — Semantic Retrieval & Whisper Voice Input

Dense Semantic Retrieval with Sentence-BERT: Unlike fragile keyword search, the system encodes the department FAQ database using all-MiniLM-L6-v2 into 384-dimensional dense vectors. During query inference, the user prompt is projected into the same latent space to compute cosine similarities against both question and answer representations:

$$\text{Sim}(q, d) = (u \cdot v) / (||u|| \, ||v||)$$

A hybrid lexical boost adds up to +0.20 for exact keyword overlaps (e.g., 'fees', 'attendance', 'hod', 'gpu'). Queries falling below the threshold ($\theta = 0.35$) are flagged as out-of-domain or ungrounded.

OpenAI Whisper Speech-to-Text Pipeline: Spoken queries recorded via microphone or uploaded audio files are fed into OpenAI Whisper (base model). Audio is normalized, resampled to 16,000 Hz 16-bit mono PCM, and transcribed with zero external API dependencies. The resulting text seamlessly integrates into the multi-turn conversational memory.

5. Unit U4 — TTS Voice Synthesis & Event Poster Studio

Text-to-Speech (TTS) Voice Engine: Responses are converted to natural spoken audio via Google Text-to-Speech (gTTS) with an offline fallback to macOS native speech synthesizers. The TTS pre-processor cleans markdown asterisks, URLs, and code brackets to ensure smooth, pleasant auditory delivery.

Automated Publication Poster Studio: When users inquire about departmental events (e.g., *AI-Conflux Hackathon* or *NeurOlympics*) or request promotional flyers, the high-resolution Pillow graphic engine renders a 1080×1440 publication poster. It features cybernetic grid overlays, typography hierarchies, highlight badges, venue cards, and a procedural registration QR code.

Figure 2: Publication-Ready Event Poster Generated for Department Hackathon



6. Unit U5 — Safety Guardrails, Hallucination Blocking & Zero-PII Policy

To ensure departmental compliance, strict three-tiered guardrails are enforced on every interaction:

- **Out-of-Scope Query Refusal:** Unrelated queries (politics, cooking recipes, sports, entertainment, personal counseling) are intercepted by regex classifiers and semantic distance checks, triggering an official academic refusal.
- **Hallucination & Unsupported Answer Blocking:** If an in-domain question lacks grounding documentation in the FAQ database (confidence < 0.35), the chatbot is strictly prevented from guessing or fabricating dates/numbers. It returns an official referral to Room 304.
- **Zero Personal Data Retention (Anti-PII):** User inputs containing phone numbers, emails, credit cards, or government IDs are scrubbed in real-time (replaced with [REDACTED_PHONE], [REDACTED_EMAIL]). No user data is ever written to disk.

7. Comprehensive Test Suite Validation Matrix

Test Case / Unit	Input Query / Action	Expected Behavior	Validation Status
U1: System Prompt	Inspect get_system_prompt()	Contains persona, boundaries & tone	■ PASS (Verified)
U2: Multi-Turn Memory	2-turn sequential dialog	Preserves history across sliding buffer	■ PASS (Verified)
U2: LoRA Convergence	4-Epoch PEFT training	Loss drops 9.81 -> 6.17; PPL 18k -> 480	■ PASS (Verified)
U3: Semantic Retrieval	'How much attendance needed?'	Cosine similarity > 0.50 for Doc #3	■ PASS (Score: 0.555)
U3: Whisper STT	16kHz Speech Input	Transcribes clear voice prompt	■ PASS (Verified)
U4: TTS Spoken Audio	Synthesize department greeting	Generates playable .mp3 audio file	■ PASS (Verified)
U4: Poster Generator	AI-Conflux Hackathon details	Outputs 1080x1440 PNG graphic poster	■ PASS (Verified)
U5: Out-of-Scope Refusal	'How to bake a chocolate cake?'	Polite refusal & scope clarification	■ PASS (Blocked)
U5: Anti-PII Redaction	Input: Phone & Email credentials	Replaces with [REDACTED_*] tokens	■ PASS (Zero PII Saved)
U5: Hallucination Blocker	Ungrounded / made-up query	Refuses fabrication; directs to HOD	■ PASS (Blocked)

8. Live Application Universal URL & Deployment Guide

The project is structured for immediate worldwide access from any browser without complex firewall configurations:

- **Universal Live App URL (Gradio Secure FRP Tunnel):**
  https://fac2e8541ef1e9a672.gradio.live

To launch a new live session at any time: python3 app.py --share
Gradio instantly binds to the local backend, registers an encrypted SSL public gateway, and yields a universal live link.

- **Permanent Cloud Mirror (Hugging Face Spaces):**
  https://huggingface.co/spaces/jeetdoesthings/FAQChatbot

- **Official GitHub Repository:**
  https://github.com/jeetdoesthings/FAQChatbot

Contains complete source code, LoRA adapter weights, dataset JSON, test scripts, and demo instructions.

9. Conclusion

Project P24 successfully demonstrates a production-caliber Department FAQ Voice Chatbot satisfying all units U1 to U5. Through the fusion of dense Sentence-BERT retrieval, OpenAI Whisper speech recognition, LoRA parameter-efficient adaptation, speech synthesis, graphic design automation, and multi-layered safety guardrails, the chatbot delivers a responsive, reliable, and privacy-preserving virtual assistant for higher education departments.